

AI exercise: read the regression table with an AI

You will mark a written AI response, then run an improved prompt in Microsoft Copilot and compare the two. Keep page 1 of your worksheet (the regression table) and your AI good-practice card beside you. The whole exercise takes about 15 minutes.

GreenWaste is a fictional training case.

The table you are reading (a copy of the table on your worksheet)

Result row	Coefficient AED	95% interval AED	p-value
Before GreenWaste	1,432	1,418 to 1,446	< 0.001
After minus before	-669	-684 to -654	< 0.001

Part A: Mark the response (about 8 minutes)

Naive prompt (a prompt with no context, as a busy person might type it): "Explain this table and advise whether to expand GreenWaste." This response was written for the exercise. It is not a real AI answer.

The city data show a substantial improvement in participant businesses' annual waste management cost: the mean fell from 1,432 AED before GreenWaste to 763 AED twelve months later. The reported change is -669 AED, with a 95% confidence interval of -684 to -654 AED and $p < 0.001$. This is strong statistical evidence against a zero mean change.

Because the same businesses were measured twice, their starting differences are accounted for, so the programme can be credited with this reduction. The narrow interval suggests the finding is robust, and most individual businesses can expect savings in this range.

The recorded reduction is below our rule of at least a 1,000 AED reduction, but the very small p-value makes a compelling case for expansion. A phased rollout with monitoring would be prudent to confirm the result in other locations.

Underline the claims that the regression table supports. Cross out the claims that it does not support. Use the table, not how convincing the paragraph sounds.

Which claim would you correct first, and what would you write instead?

Part B: Run the improved prompt, then compare

Copy the prompt below into Microsoft Copilot. The table is included in the prompt. If you are not using Copilot, read the prepared response below after you finish Part A.

Improved prompt

Help me read a GreenWaste before-and-after regression table for a director. The outcome is annual waste management cost in AED per business, measured before and 12 months after GreenWaste. The same 4,794 participant businesses were measured twice. No comparison businesses that did not receive GreenWaste are included. Our rule is a reduction in annual waste management cost of at least 1,000 AED per business per year.

The table (annual waste management cost in AED per business). Before GreenWaste: coefficient 1,432, 95% interval 1,418 to 1,446, $p < 0.001$. After minus before: coefficient -669, 95% interval -684 to -654, $p < 0.001$.

Ask me any essential clarifying questions before you recommend action. Separate the recorded change, the statistical uncertainty, the question of whether GreenWaste caused the change, and the importance of the change for our decision. Explain the coefficient, the confidence interval and the p-value in plain language. Cite the table for factual claims. Do not invent missing evidence, and do not assume the interval describes individual businesses.

After the AI has answered, write your comparison.

(a) Which question that the AI asked helped you most?

(b) What improved compared with the response in Part A?

(c) Which claim still needs checking against the table?

A better prompt reduces mistakes. It does not remove the need to check every number and claim against the table.

Prepared response for participants not using Copilot

Read this after you have marked the response in Part A. It is an example of a better answer. An AI tool may answer differently. This response was written for the exercise. It is not a real AI answer.

Before I advise on expansion, I need to know what happened to comparison businesses that did not receive GreenWaste, and whether our rule of at least a 1,000 AED reduction is the only criterion for the decision.

The table reports a mean change in annual waste management cost of -669 AED among these city participant businesses over 12 months, with a 95% interval of -684 to -654 AED. This is a statistically significant recorded fall under the model, but a before-and-after comparison alone does not show what GreenWaste caused. The interval describes the estimated mean change. It does not describe the savings of individual businesses, and it does not allow for every source of bias.

The p-value starts from the assumption that the mean change is zero. It does not show that GreenWaste caused the fall, and it does not show that the change is large enough to matter. The recorded reduction and its interval are below 1,000 AED. That does not show that GreenWaste's causal effect is below our rule. I would request evidence from comparison businesses before attributing the fall to GreenWaste, and evidence on the full programme costs before recommending expansion.

(a) Which sentences does the regression table support?

(b) What additional evidence would answer the clarifying question at the start of the response?
